

Function Pointers

- A pointer to a function contains the address of the function in memory.
- The name of an array is actually the address in memory of the first element of the array.
- The name of a function is actually the starting address in memory of the code that performs the function's task.

```
int main()
{
    const int arraySize = 10;
    int choice; // 1 = minimum, 2 = maximum
    int data[ arraySize ] = { 2, 6, 4, 8, 10, 12, 89, 68, 45, 37 };

    for( int counter = 0; counter < arraySize; counter++ )
        cout << setw( 4 ) << data[ counter ];

    cout << "\n\nEnter 1 to find minimum, Enter 2 to find maximum: ";
    cin >> choice;
    cout << endl;

    if( choice == 1 )
    {
        int minimum = extremeValue( data, arraySize, greater );
        cout << "The minimum: " << minimum << endl << endl;
    }
    else
    {
        int maximum = extremeValue( data, arraySize, less );
        cout << "The maximum: " << maximum << endl << endl;
    }
}
```

```
int extremeValue( int data[], const int size,  
                  bool ( *compare )( int, int ) )  
{  
    int minOrMax = data[ 0 ];  
    for( int i = 0; i < size - 1; i++ )  
        if( ( *compare )( minOrMax, data[ i ] ) )  
            minOrMax = data[ i ];  
  
    return minOrMax;  
}
```

// returns true if a is less than b

```
bool less( int a, int b )  
{  
    return a < b;  
}
```

// returns true if a is greater than b

```
bool greater( int a, int b )  
{  
    return a > b;  
}
```

Data items:

2 6 4 8 10 12 89 68 45 37

Enter 1 to find minimum, Enter 2 to find maximum: 1

The minimum: 2

Data items:

2 6 4 8 10 12 89 68 45 37

Enter 1 to find minimum, Enter 2 to find maximum: 2

The maximum: 89